Master EIT ICT Labs in ICT Innovation

Universidad Politécnica de Madrid

Non textual information retrieval

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D-2315

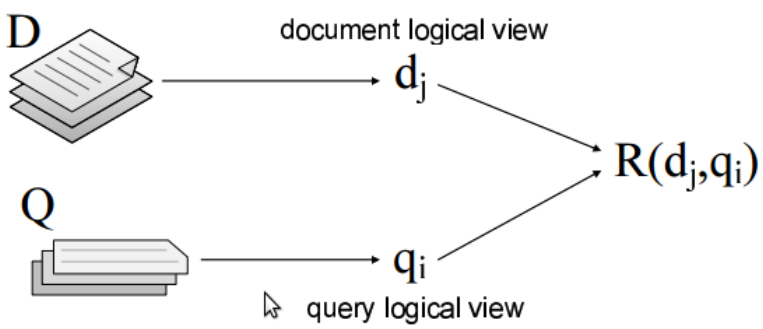
Which kind of data

- Multimedia information retrieval
  - Image
  - Audio
  - Videos
  - Other signals

- IR - Huge research area
  - We will have a brief survey on the field

IR Models

- An **IR model** is a quadruple $[\mathbf{D}, \mathbf{Q}, \mathcal{F}, R(q_i, d_j)]$ where
  1. $\mathbf{D}$ is a set of logical views for the documents in the collection
  2. $\mathbf{Q}$ is a set of logical views for the user queries
  3. $\mathcal{F}$ is a framework for modeling documents and queries
  4. $R(q_i, d_j)$ is a ranking function



How we can extract information

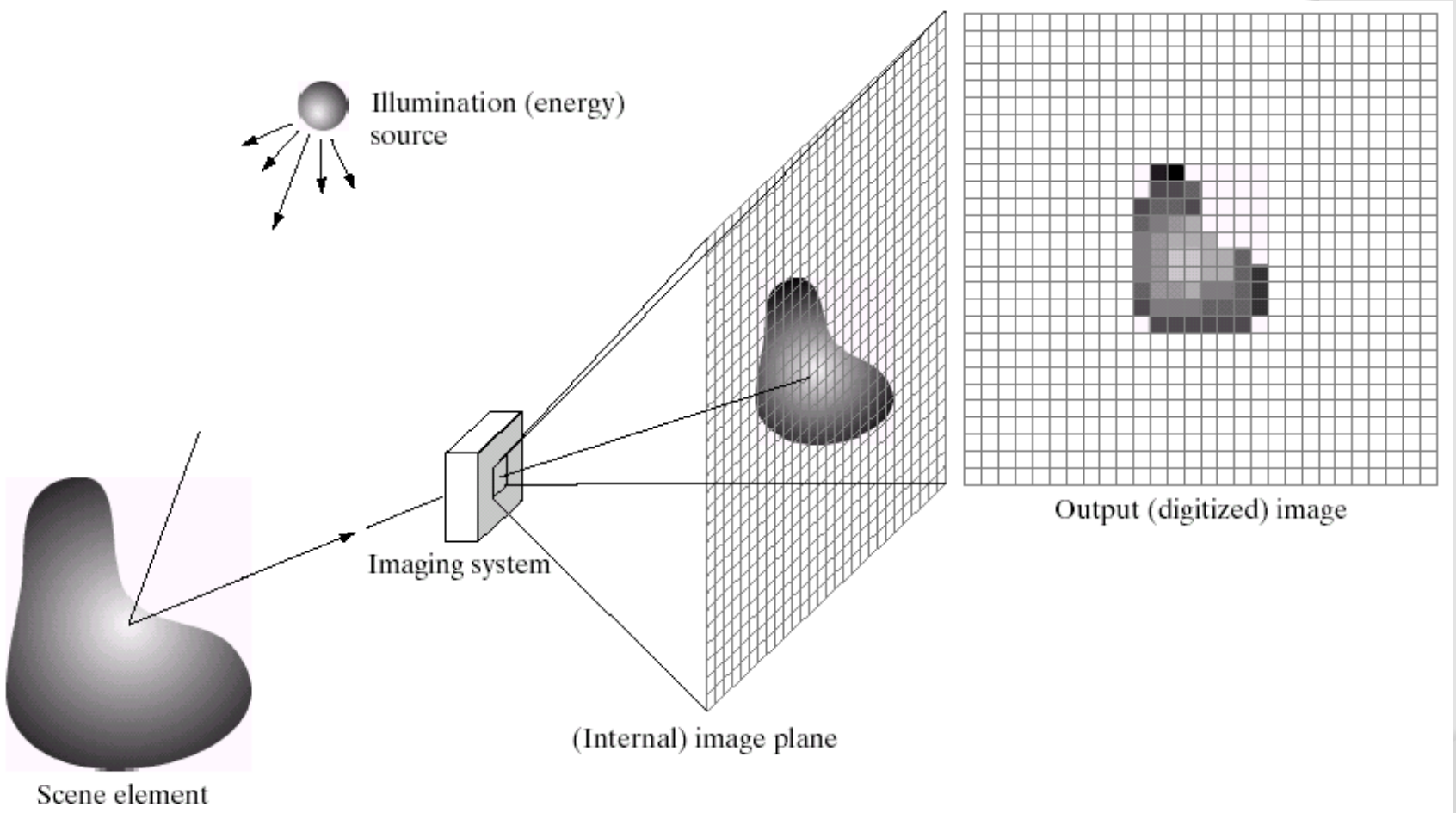
- Logical view for non textual data
- Ranking function for comparing data and queries

Which kind of data

- Multimedia ??
  - Image: OCR, fingerprint and face recognition
  - Audio: Music Information Retrieval, voice recognition
  - Videos: Shot slicing, enities
  - Signals: devices (pacemaker, ...), and sensors(gyroscope, ...)

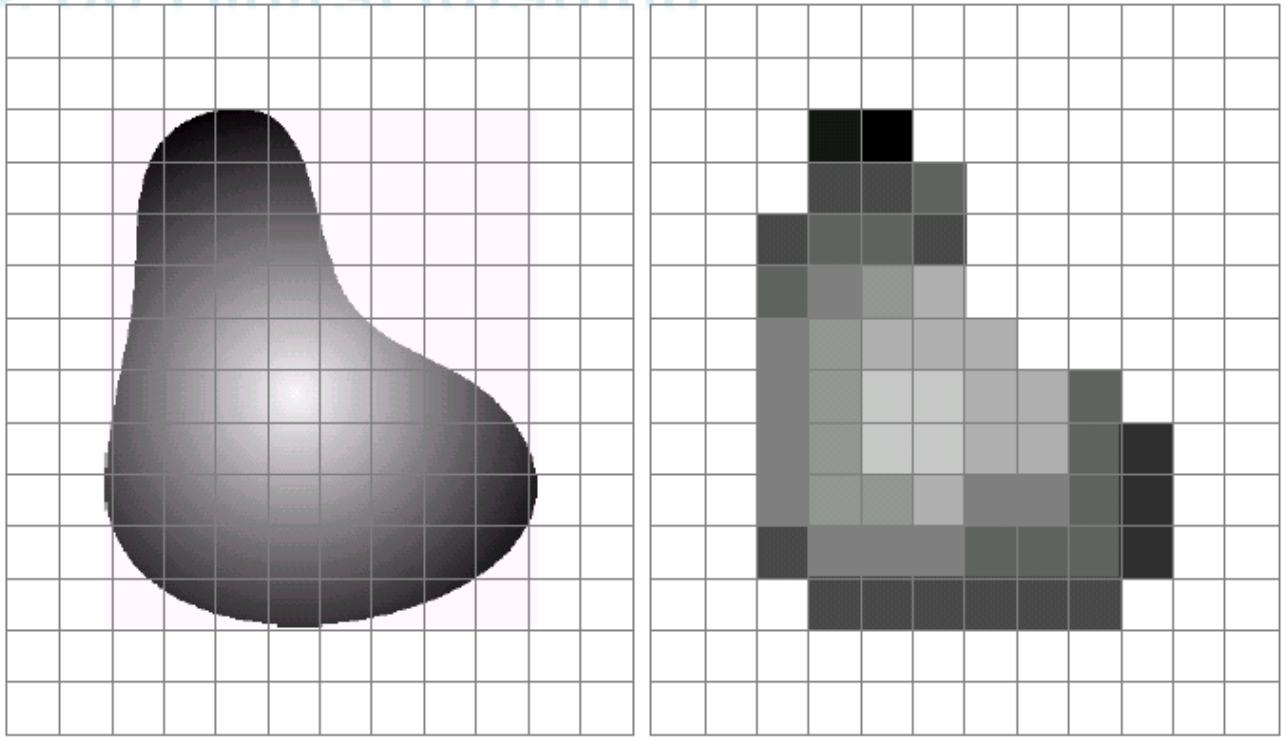
Very brief introduction to image processing

Basics on Digital imaging



(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Basics on Digital imaging



a b

FIGURE 2.17 (a) Continuos image projected onto a sensor array. (b) Result of image sampling and quantization.

Image sampling: discretize an image in the spatial domain

Spatial resolution / image resolution: pixel size or number of pixels

(Images from Rafael C. Gonzalez and Richard E
Wood, Digital Image Processing, 2nd Edition.

Basics on Digital imaging

- Image quantization:
  - discretize continuous pixel values into discrete numbers
- Color resolution/ color depth/ levels:
  - Number of colors or gray levels
  - Number of bits representing each pixel value

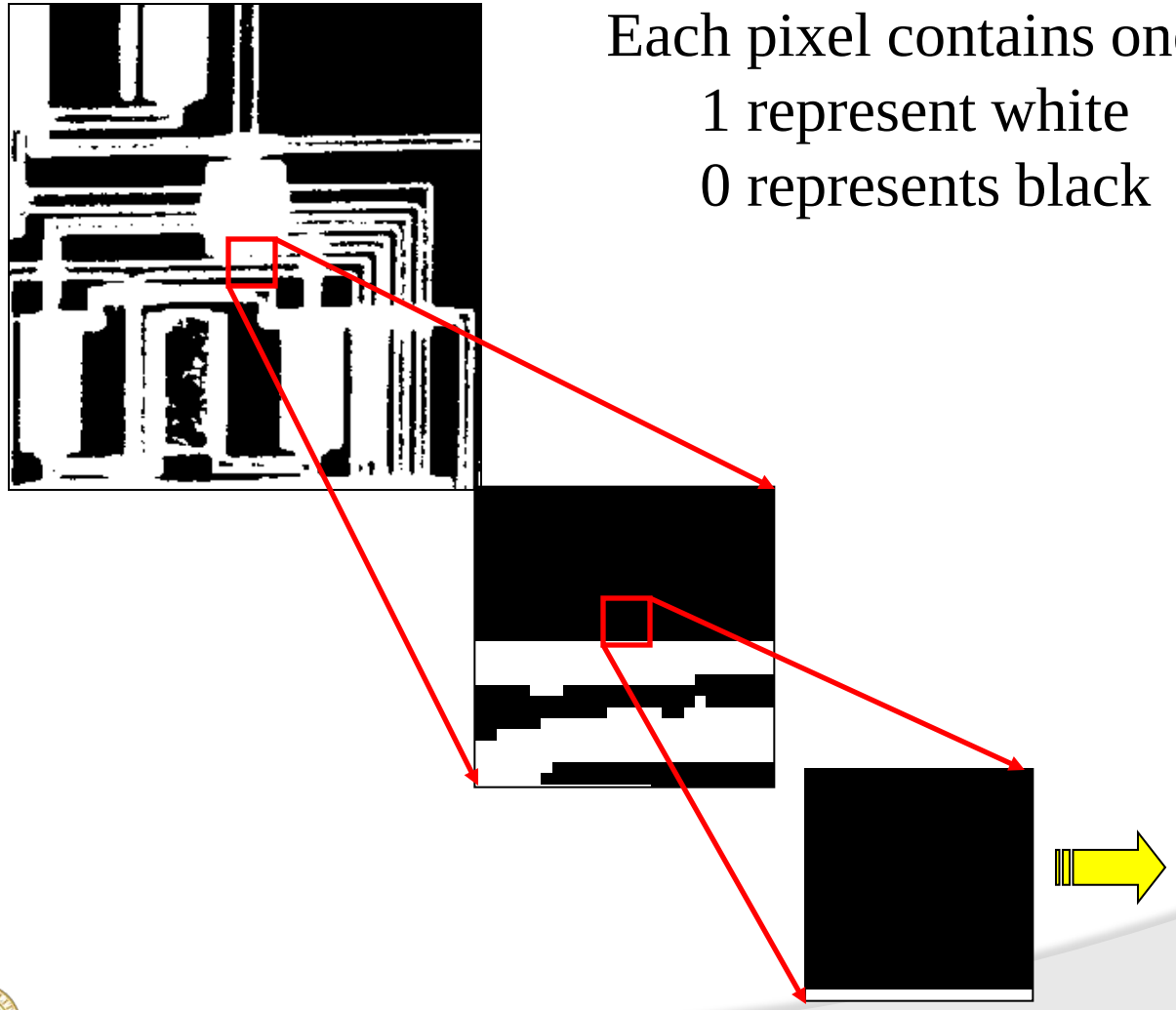
Basics on Digital imaging

Binary image or black and white image

Each pixel contains one bit :

1 represent white

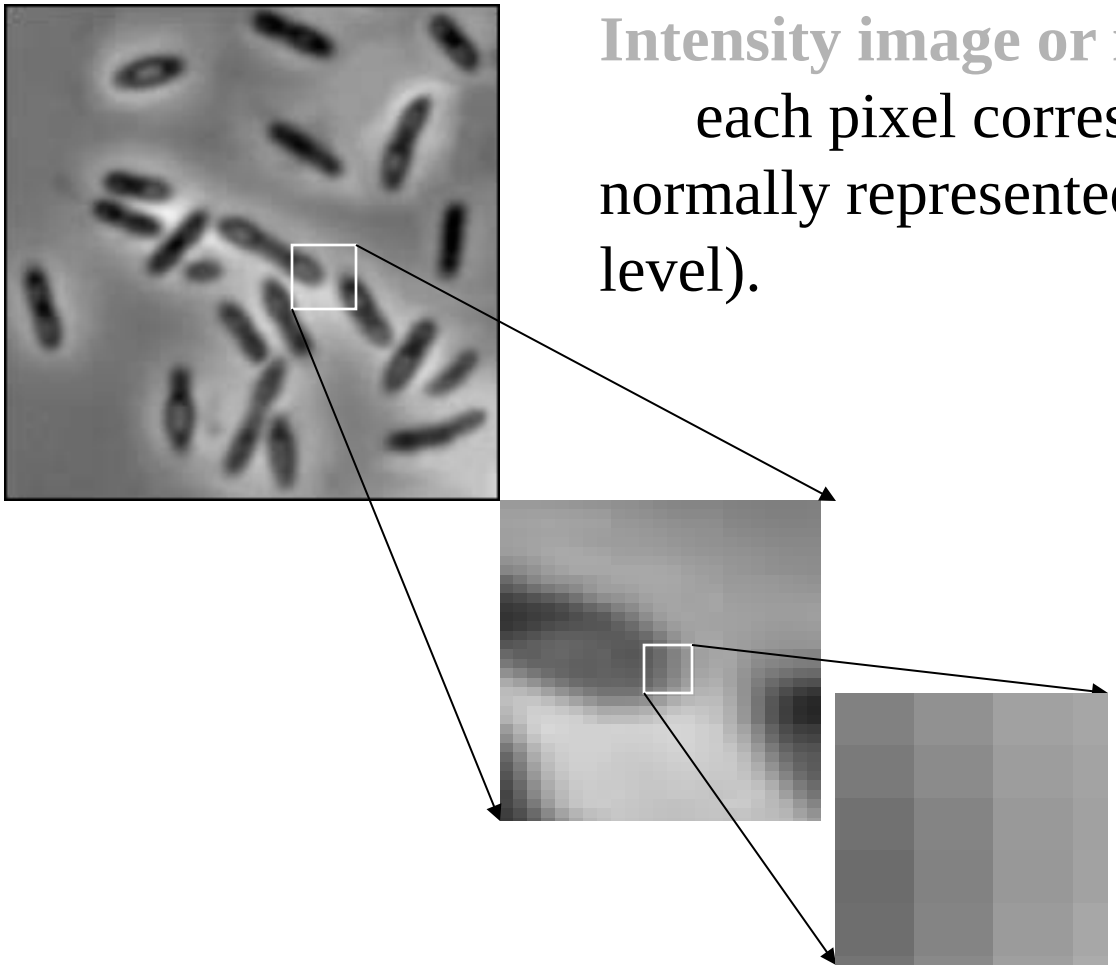
0 represents black



Binary data

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Basics on Digital imaging



Intensity image or monochrome image

each pixel corresponds to light intensity normally represented in gray scale (gray level).

Gray scale values

$$\begin{bmatrix} 10 & 10 & 16 & 28 \\ 9 & 6 & 26 & 37 \\ 15 & 25 & 13 & 22 \\ 32 & 15 & 87 & 39 \end{bmatrix}$$

Basics on Digital imaging

Color image or RGB image: each pixel contains a vector representing red, green and blue components.

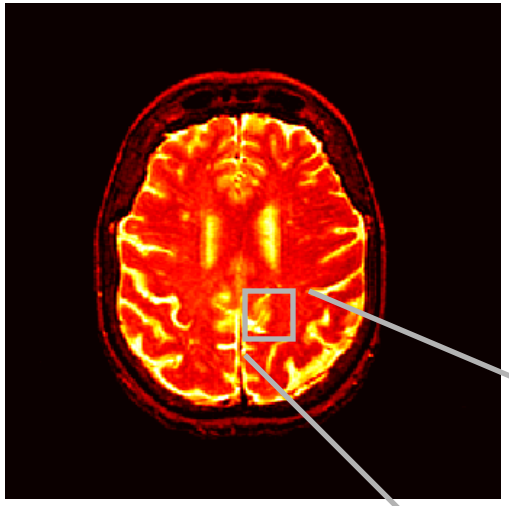
RGB components

$$\begin{bmatrix} 10 & 10 & 16 & 28 \\ 9 & \\ 15 & \\ 32 & \end{bmatrix} \begin{bmatrix} 65 & 70 & 56 & 43 \\ 32 & \\ 21 & \\ 54 & \end{bmatrix} \begin{bmatrix} 99 & 70 & 56 & 78 \\ 60 & 90 & 96 & 67 \\ 85 & 85 & 43 & 92 \\ 32 & 65 & 87 & 99 \end{bmatrix}$$

Basics on Digital imaging

Index image

Each pixel contains index number pointing to a color in a color table



$$\begin{bmatrix} 1 & 4 & 9 \\ 6 & 4 & 7 \\ 6 & 5 & 2 \end{bmatrix}$$

Index value

Color Table

Index No.	Red component	Green component	Blue component
1	0.1	0.5	0.3
2	1.0	0.0	0.0
3	0.0	1.0	0.0
4	0.5	0.5	0.5
5	0.2	0.8	0.9
…	…	…	…

Basics on Digital imaging

Image “After snow storm”

Image: multidimensional function of spatial coordinates.

Coordinates: 2D - photograph, 3D CT scan or movies

Values: intensity or color values

What are we searching in our images?

How we can extract information

- Logical view for non textual data
  - Objects, descriptors, basic features
- Ranking function for comparing data and queries
  - Distance function, comparator, matcher

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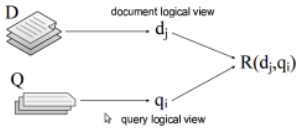


3

Chap 03: Modeling, Baeza-Yates & Ribeiro-Neto, Modern Information Retrieval, 2nd Edition – p. 6

Overview

- Image description: characteristics
  - **Describe the entire image by summarizing it**
  - Describe image by its objects / components
  - Describe image by Keypoints
  - Describe image contents by textual notes: metadata

Describe the entire image

- Feature extraction
  - Use the most fundamental elements an image consists of
  - Low-level features are the lowest, the most basic features of an image
  - Typically divided in three different :
    - Color
    - Shape
    - Texture.

Describe the entire image: Color

- It is the easiest method

- Each pixel in a digital image consist of a color element.
  - In a grey-scale image:
    - Range from 0 to 255, (from 0 black to 255 white)
  - In a color image (24-bit color)
    - Divided in three color components RGB
    - 8 bits is used to represent each component
    - (from 0 black to 255 red/green/blue)

Describe the entire image: Color histogram

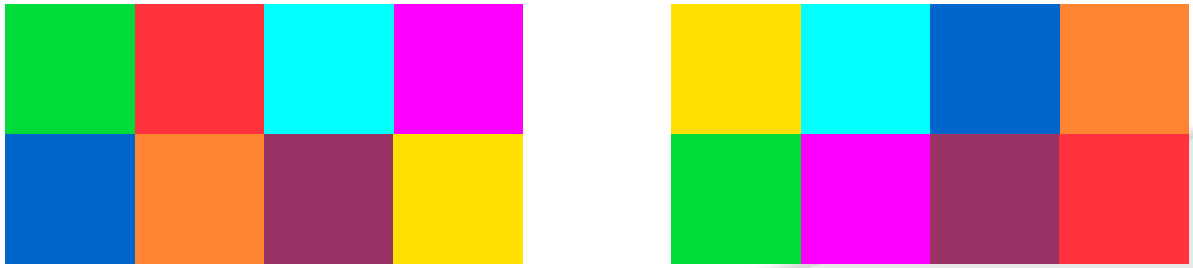
- Typical way of indexing color is using color histograms
- A color histogram is a table with different color values as the index key, and the sum of the occurrences of these colors as the value
- Pixel labels can be grouped as bins

- How can we use this histogram in order to retrieve other, similar images?
- The most basic technique is to subtract the query histogram color-for-color with each of the histograms in the image collection

Describe the entire image: Color histogram

- Common problems with histograms
  - Small differences in color distribution

  - Histograms tend to favor background
  - Differences in spatial color distribution



Describe the entire image: Color histogram

- Solution to differences in color distribution
  - Equalize intensities in the image
  - Histogram equalization – try to normalize intensity

- Solution to differences in spatial color distribution
  - Perceptually weighted histogram (PWH), pre-processing by dividing the image into several pieces building a histogram of each of the pieces
  - Part-histograms are compared to the corresponding in other image

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Describe components in the image

- Segmentation of the original image in objects
- These objects try to corresponds to real objects perceived by humans

- Descriptors are calculated for each region
  - Centroid
  - Perimeter
  - Color descriptors
  - Texture descriptors

- Comparison is object-based

Describe components in the image: Segmentation

- Segmentation partitions an image into distinct regions containing each pixels with similar properties

- Regions should strongly relate to depicted objects or features of interest
  - For interpreting correctly an image, a meaningful segmentation transforming image into terms of features, objects, and scenes

- The success of image analysis depends on reliability of segmentation, but an accurate partitioning of an image is generally a very challenging problem

Blob detection / contours and describe them

- From J. Russ Image Processing Handbook, Vol. 2, p. 525, there are some definitions of the blob

- Formfactor 4 pi Area / Perimeter^2
- Roundness 4 Area / (pi Max-diam^2)
- Aspect ratio Max-diam / Min-diam
- Curl max caliper diam / fiber length
- Convexity Convex Perimeter / Perimeter
- Solidity Area / Convex Area
- Compactness sqrt (4 * area / pi) / max diameter
- Modification ratio inscribed diameter / max diameter
- Extent Net Area / area of bounding rectangle

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Describe image by Keypoints

- Not aim to divide image in regions or images
- Select some points for describe the image
- Directed / Ad-hoc feature descriptors
  - Fingerprint recognition
  - Facial recognition
  - OCR
  - Automatic number plate recognition
- Implies knowledge about image
  - SIFT / SURF
  - FAST
  - ORB
  - …

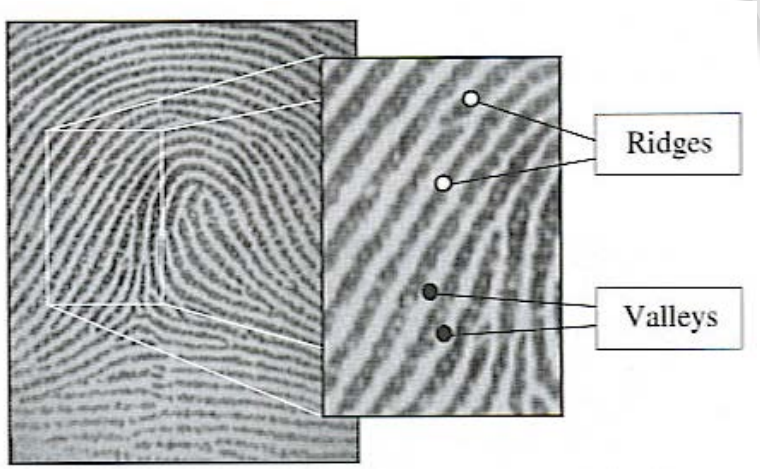
Fingerprint detection

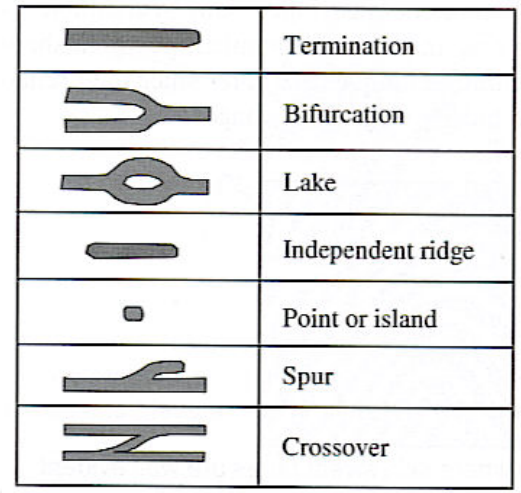
The lines that flow in various patterns across fingerprints are called **ridges** and the space between ridges are **valleys**.

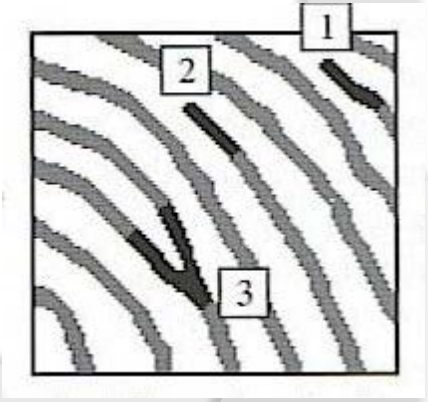
Fingerprint features (associated with some matching algorithm):

Ridge pattern - global pattern matching;

minutiae (ridge ending and ridge bifurcation)
- minutiae matching;
- attributes: type, (x,y)
- location, orientation







1 and 2 are endings; 3 is bifurcation

Facial recognition

- Traditional facial recognition extracts features from an image of the subject's face relative position
  - Size
  - Shape of the eyes
  - Nose
  - Cheekbones
  - Jaw
- These features are then used to search for other images with matching features

OCR

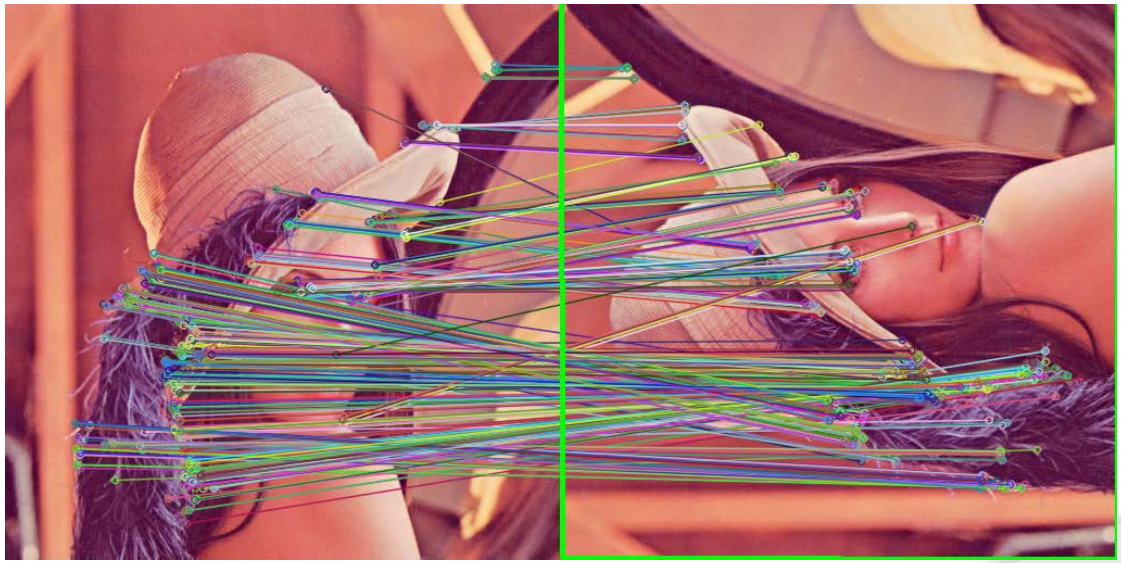
- Optical Character Recognition, or OCR, is a technology that enables you to convert different types of documents, such as scanned paper documents, PDF files or images captured by a digital camera into editable and searchable data

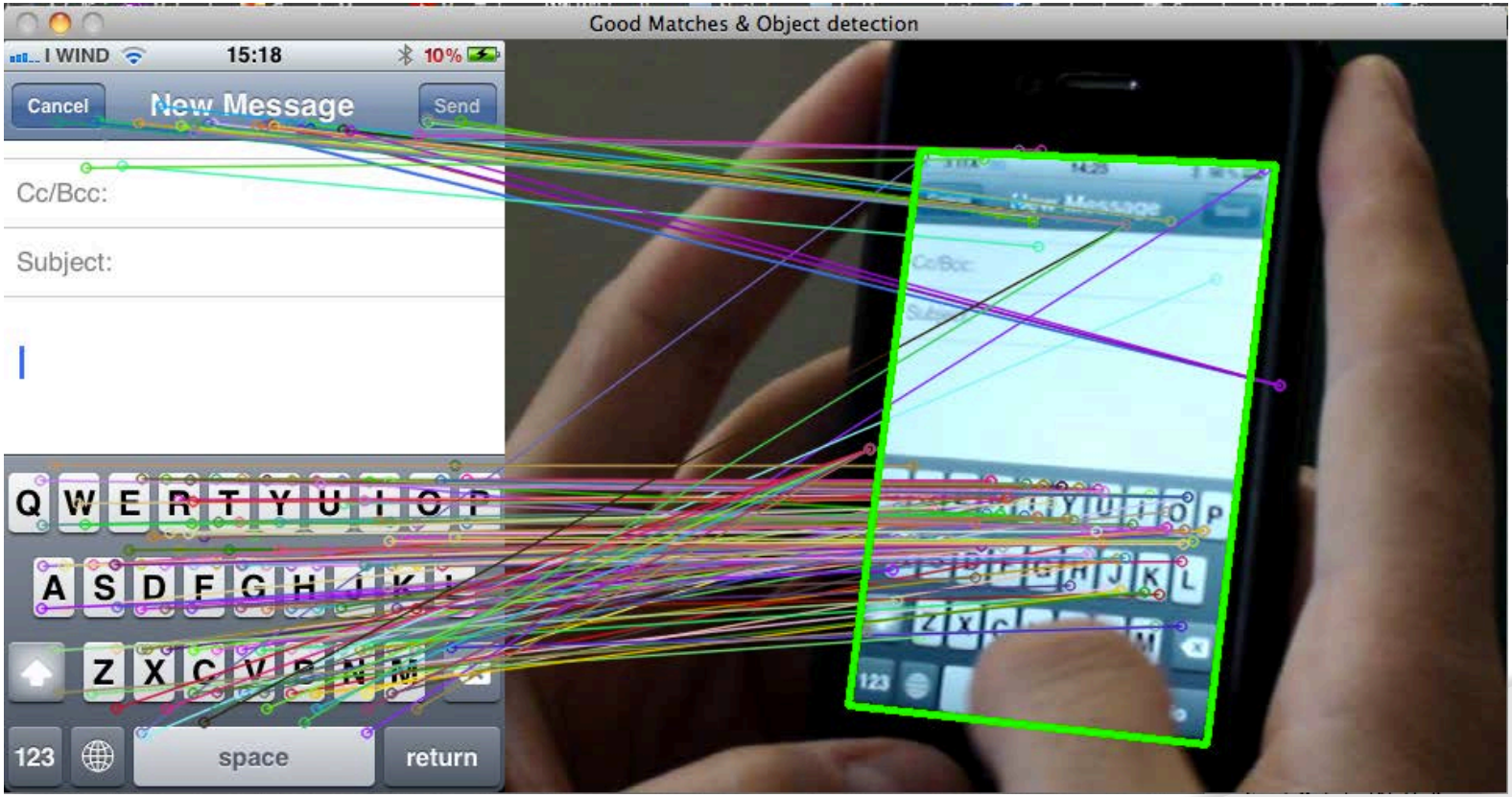
Describe image by Keypoints

- Algorithms for detecting generic keypoints and describing
  - SIFT / SURF
  - FAST
  - ORB
  - BRIEF
  - FREAK
  - …
- Often costly matching
- 1:1 comparison

Describe image by Keypoints

- Desiderable: Invariance to size and orientation
- Matching of keypoints example (ORB+BFM)




Good Matches & Object detection
I WIND
15:18
10%
Cancel
New Message
Send
Cc/Bcc:
Subject:
space
return
123

Overview

- Image description: characteristics
  - Describe the entire image by summarizing it
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  - **Describe image contents by textual notes: metadata**

Describe image contents: metadata

- EXIF
- Dublin Core Metadata
- Standard MPEG7
- DICOM standard

DCM: Dublin Core Metadata

<metadata xmlns="http://w.x.y/z/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://w.x.y/z/ http://w.x.y/z/schema.xsd"
xmlns:dc="http://purl.org/dc/elements/1.1/">

<dc:title> **Multimedia information retrieval** </dc:title>

<dc:description>

This presentation covers the process of indexing and finding multimedia documents through …

</dc:description>

<dc:creator> **Raul Alonso** </dc:creator>

<dc:date> **2016** </dc:date>

<dc:publisher> **UPM** </dc:publisher>

<dc:identifier> **ISBN paper …** </dc:identifier>

<dc:identifier> **ISBN ebook …** </dc:identifier>

</metadata>

MPEG 7

- Most salient metadata standard for multimedia is MPEG-7 for description and search of audiovisual resources
- Developed by MPEG (Moving Picture Experts Group)
- Also contains low-level descriptors that can be used with content-based queries
- Also proposes a set of features that can be used for this purpose
  - but the standard does not suggest nor prescribe how content-based queries should be carried out.

The Parts of the DICOM Standard

- Part 1 - Introduction and Overview
- Part 2 - Conformance
- Part 3 - Information Object Definitions
- Part 4 - Service Class Definitions
- Part 5 - Data Structures & Semantics
- Part 6 - Data Element Listing and Typing
- Part 7 - Message Exchange Protocol
- Part 8 - Network Support for Message Exchange
- Part 9 - Point-to-Point Support

The Parts of the DICOM Standard

- Part 10 - Media Storage and File Format
- Part 11 - Media Storage Application Profiles
- Part 12 - Media Formats and Physical Media
- Part 13 - Print Management Point-to-Point

References

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- http://slideplayer.com/slide/4512105/
- http://www.imageprocessingplace.com/root_files_V3/image_databases.htm

- https://en.wikipedia.org/wiki/Automatic_number_plate_recognition
- https://en.wikipedia.org/wiki/Facial_recognition_system